

Jackknife Inference for Fixed Effects Models

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Overview

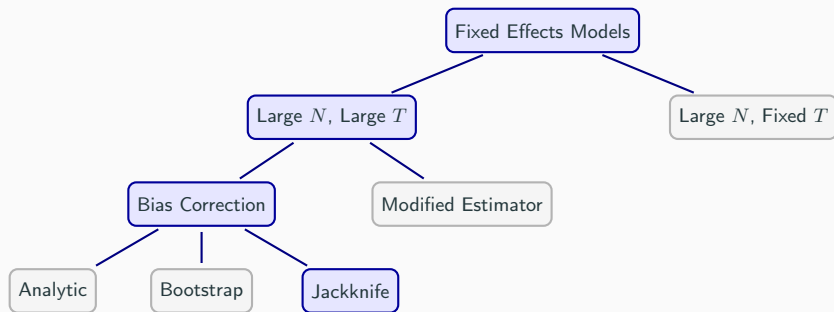


Figure 1: Overview.

Fixed Effects Models

- Fixed effects models are commonly used in economics.
- They are used to control for certain types of endogeneity.
- Example:

$$\text{wage}_{it} = \beta_0 + \beta_1 \text{educ}_{it} + \beta_2 \text{ability}_i + \varepsilon_{it},$$

with $\mathbb{E}[\varepsilon_{it} | \text{educ}_{it}, \text{ability}_i] = 0$.

- How do expected wages respond to a change in educ_{it} , holding ability_i fixed?

$$\frac{\partial \mathbb{E}[\text{wage}_{it} | \text{educ}_{it}, \text{ability}_i]}{\partial \text{educ}_{it}} = \beta_1.$$

Fixed Effects Models

- Assume that ability_i unobserved.
- Since ability_i influences educ_{it} , we observe correlation between the two variables.
- Consequently,

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 + \beta_2 \frac{\text{Cov}(\text{educ}_{it}, \text{ability}_i)}{\text{Var}(\text{educ}_{it})},$$

i.e., LS picks up the wrong effect.

Fixed Effects Models

- A (possible) solution is to treat $\beta_2 \text{ ability}_i = \lambda_i$ as a new parameter and estimate

$$\text{wage}_{it} = \beta_0 + \beta_1 \text{educ}_{it} + \lambda_i + \varepsilon_{it}.$$

- This is known as the fixed effects approach (conditioning on ability) and can allow us to estimate β_1 .
- However, introducing individual specific parameters leads to other issues.

Fixed Effects Models

- Large N , fixed T :
 - Rarely are standard estimators consistent. (e.g., linear regression with strictly exogenous regressors)
 - Sometimes possible to construct alternative consistent estimators. (e.g., dynamic linear regression)
 - Sometimes the parameter of interest is not even identified with small T . (e.g., dynamic binary outcome)
- Large N , large T :
 - Standard estimators are often consistent.
 - They suffer from asymptotic bias.

- Typically for fixed effects estimators

$$\sqrt{NT}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(\mu, \sigma^2),$$

as $N, T \rightarrow \infty$ and $N/T \rightarrow \gamma^2 \in (0, \infty)$

- μ referred to as asymptotic bias.
- Standard inference procedures (LR tests, Wald tests, etc) are based on (asymptotically) unbiased statistics.

Bias Correction

- Analytic bias correction:
 - Must be established on a case-by-case basis.
 - Complex formula, especially for functions of parameters.
 - Tuning parameters.
 - Increasingly hard to establish.
- Jackknife bias correction:
 - Delete-one-jackknife.
 - Split-panel jackknife.
 - Only semi-automatic.
- Bootstrap “bias correction”:
 - Does not always work.
 - Computational cost.

This Paper

In this paper I return to the jackknife and develop this into a complete, fully-automatic inference procedure.

Assumption AD: There exists an estimator of φ , denoted by $\hat{\varphi}^{(0)}$, such that

$$r_{NT}(\hat{\varphi}^{(0)} - \varphi) = z_{NT} + \sum_{r=1}^R \mu_{r,NT} + \mathcal{O}_p(1),$$

as $N, T \rightarrow \infty$ and $N/T \rightarrow \gamma^2 \in (0, \infty)$, where $\mu_{1,NT}, \dots, \mu_{R,NT}$ are nonstochastic, but possibly N and T dependent, and

$$z_{NT} \xrightarrow{d} \mathcal{N}(0, \sigma^2).$$

Assumption AD

- Expansions of this form have been established for a large range of models, estimators, and parameters of interest.
 - LS and GMM estimators of linear dynamic panel models with individual fixed effects ([Nickell \(1981\)](#), [Hahn and Kuersteiner \(2002\)](#), [Alvarez and Arellano \(2003\)](#))
 - Extremum estimators of nonlinear models with individual fixed effects such as logit and probit models ([Hahn and Newey \(2004\)](#), [Fernández-Val \(2009\)](#), [Hahn and Kuersteiner \(2011\)](#))
 - Nonlinear models with individual and time fixed effects ([Fernández-Val and Weidner \(2016\)](#))
 - LS and ML estimators of linear and nonlinear models with interactive fixed effects ([Bai \(2009\)](#), [Moon and Weidner \(2017\)](#), [Chen, Fernández-Val, and Weidner \(2021\)](#)) ...

Example

Outcomes are generated according to

$$y_{it} = \lambda_i + \varphi y_{i,t-1} + \varepsilon_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T,$$

where $\varepsilon_{it} \sim \text{iid } \mathcal{N}(0, 1)$, ε_{it} is independent of λ_i , and $|\varphi| < 1$. Let $\hat{\varphi}^{(0)}$ be the LS estimator of φ . As $N, T \rightarrow \infty$ and $N/T \rightarrow \gamma^2 \in (0, \infty)$,

$$\sqrt{NT}(\hat{\varphi}^{(0)} - \varphi) = \frac{(1 - \varphi^2)}{\sqrt{NT}} \sum_{(i,t) \in S_0} y_{i,t-1} \varepsilon_{it} - \sqrt{\frac{N}{T}}(1 + \varphi) + o_p(1),$$

where $S_0 := \{1, \dots, N\} \times \{1, \dots, T\}$, and

$$\frac{(1 - \varphi^2)}{\sqrt{NT}} \sum_{(i,t) \in S_0} y_{i,t-1} \varepsilon_{it} \xrightarrow{d} \mathcal{N}(0, 1 - \varphi^2).$$

Example

Hence,

$$\underbrace{\sqrt{NT}}_{r_{NT}} (\hat{\varphi}^{(0)} - \varphi) = \underbrace{\frac{(1 - \varphi^2)}{\sqrt{NT}} \sum_{(i,t) \in S_0} y_{i,t-1} \varepsilon_{it}}_{z_{NT}} + \underbrace{-\sqrt{\frac{N}{T}}(1 + \varphi)}_{\mu_{NT}} + \mathcal{O}_p(1),$$

$R = 1$ and $\sigma^2 = 1 - \varphi^2$.

Assumption JK: Let $m \geq R + 2$ be a fixed integer. There exist $m - 1$ estimators $\hat{\varphi}^{(1)}, \dots, \hat{\varphi}^{(m-1)}$ obtained from subsamples of the data such that

$$r_{NT}(\hat{\varphi} - \varphi \boldsymbol{\iota}_m) = \mathbf{z}_{NT} + \mathbf{A}\boldsymbol{\mu}_{NT} + \mathcal{O}_p(1),$$

as $N, T \rightarrow \infty$ and $N/T \rightarrow \gamma^2 \in (0, \infty)$, where

$$\mathbf{z}_{NT} \xrightarrow{d} \mathcal{N}(\mathbf{0}_m, \sigma^2 \mathbf{C}),$$

$\hat{\varphi} := (\hat{\varphi}^{(0)}, \dots, \hat{\varphi}^{(m-1)})^\top$, $\mathbf{A}$ and $\mathbf{C}$ are known matrices of dimension $m \times R$ and $m \times m$, respectively, $\text{rank}(\mathbf{A}) = R$, $\boldsymbol{\iota}_m \notin \text{col}(\mathbf{A})$, $\text{null}((\mathbf{A}, \boldsymbol{\iota}_m)^\top) \not\subseteq \text{null}(\mathbf{C})$, $\mathbf{C} \succeq \mathbf{0}_{m \times m}$, and $\boldsymbol{\mu}_{NT} := (\mu_{1,NT}, \dots, \mu_{R,NT})^\top$.

Example - continued

Assume that $T = 2T_2$ where T_2 is an integer. Define subsamples

$$S_1 := \{1, \dots, N\} \times \{1, \dots, T_2\},$$

$$S_2 := \{1, \dots, N\} \times \{T_2 + 1, \dots, T\}.$$

Let $\hat{\varphi}^{(j)}$ be the LS estimator using subsample S_j for $j = 1, 2$. One then obtains

$$\sqrt{NT_2}(\hat{\varphi}^{(1)} - \varphi) = \frac{1}{\sqrt{NT_2}} \sum_{(i,t) \in S_1} \nu_{it} + \sqrt{2}\mu_{NT} + \mathcal{O}_p(1),$$

and

$$\sqrt{NT_2}(\hat{\varphi}^{(2)} - \varphi) = \frac{1}{\sqrt{NT_2}} \sum_{(i,t) \in S_2} \nu_{it} + \sqrt{2}\mu_{NT} + \mathcal{O}_p(1),$$

as $N, T \rightarrow \infty$ and $N/T \rightarrow \gamma^2$.

Example - continued

Let $\hat{\varphi} := (\hat{\varphi}^{(0)}, \hat{\varphi}^{(1)}, \hat{\varphi}^{(2)})^\top$.

$$\begin{aligned} r_{NT}(\hat{\varphi} - \varphi_{\mathbf{t}3}) &= \frac{1}{\sqrt{NT}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} \sum_{(i,t) \in S_0} \nu_{it} \\ \sum_{(i,t) \in S_1} \nu_{it} \\ \sum_{(i,t) \in S_2} \nu_{it} \end{pmatrix} + \mu_{NT} \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} + \mathcal{O}_p(1) \\ &= \mathbf{z}_{NT} + \mathbf{A}\boldsymbol{\mu}_{NT} + \mathcal{O}_p(1), \end{aligned}$$

where

$$\mathbf{z}_{NT} := \frac{1}{\sqrt{NT}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} \sum_{(i,t) \in S_0} \nu_{it} \\ \sum_{(i,t) \in S_1} \nu_{it} \\ \sum_{(i,t) \in S_2} \nu_{it} \end{pmatrix}, \quad \mathbf{z}_{NT} \xrightarrow{d} \mathcal{N}(\mathbf{0}_3, \sigma^2 \mathbf{C}),$$

$$\boldsymbol{\mu}_{NT} := \mu_{NT},$$

$$\mathbf{C} := \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix}, \quad \text{and} \quad \mathbf{A} := \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

Minimum-Variance Unbiased Jackknife (MVUJ) Estimator

- Consider a vector $\mathbf{v} \in \mathbb{R}^m$ such that

$$\mathbf{v}^\top \mathbf{A} = \mathbf{0}_R^\top, \quad (1)$$

$$\mathbf{v}^\top \boldsymbol{\iota}_m = 1. \quad (2)$$

- Under Assumption JK,

$$r_{NT}(\mathbf{v}^\top \hat{\boldsymbol{\varphi}} - \varphi) \xrightarrow{d} \mathcal{N}(0, \sigma^2 \mathbf{v}^\top \mathbf{C} \mathbf{v}),$$

that is, $\mathbf{v}^\top \hat{\boldsymbol{\varphi}}$ is asymptotically unbiased.

- Minimum Euclidean norm solution:

$$\mathbf{v}^\dagger := \mathbf{D}(\mathbf{D}^\top \mathbf{D})^{-1} \mathbf{d},$$

where $\mathbf{D} := (\mathbf{A}, \boldsymbol{\iota}_m)$ and $\mathbf{d} := (\mathbf{0}_R; 1)$.

- Since we know $\mathbf{C}$ we can go further....

Minimum-Variance Unbiased Jackknife Estimator

- Let

$$\mathbf{v}^* \in \arg \min_{\mathbf{v} \in \mathcal{V}} \mathbf{v}^\top \mathbf{C} \mathbf{v}. \quad (3)$$

where $\mathcal{V} = \{\mathbf{v} \in \mathbb{R}^m : \mathbf{D}^\top \mathbf{v} = \mathbf{d}\}$.

- Under Assumptions AD and JK, the programme (3) admits at least one solution $\mathbf{v}^* \in \mathcal{V}$. (Theorem 2.1)
- Using a solution to (4), we define an MVUJ estimator as

$$\tilde{\varphi} := \mathbf{v}^{*\top} \hat{\varphi}.$$

Example - continued

Recall

$$\mathbf{A} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \quad \text{and} \quad \mathbf{C} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix}.$$

An MVUJ weights vector solves

$$\min_{\mathbf{v} \in \mathcal{V}} \mathbf{v}^\top \mathbf{C} \mathbf{v} \quad \text{where} \quad \mathcal{V} = \left\{ \mathbf{v} \in \mathbb{R}^3 : \mathbf{D}^\top \mathbf{v} = \mathbf{d} \right\},$$

with

$$\mathbf{D} := \begin{pmatrix} \mathbf{A} & \boldsymbol{\iota}_3 \end{pmatrix} \quad \text{and} \quad \mathbf{d} := \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Example - continued

Direct calculation yields (MVUJ weights unique in this case)

$$\mathbf{v}^* = \left(2, -\frac{1}{2}, -\frac{1}{2} \right)^\top.$$

Therefore the MVUJ estimator is

$$\tilde{\varphi} = \mathbf{v}^{*\top} \hat{\varphi} = 2\hat{\varphi}^{(0)} - \frac{1}{2}\hat{\varphi}^{(1)} - \frac{1}{2}\hat{\varphi}^{(2)}.$$

Moreover,

$$r_{NT}(\tilde{\varphi} - \varphi) \xrightarrow{d} \mathcal{N}(0, \sigma^2).$$

- Still need to deal with unknown variance σ^2 .
- Consider a vector $\mathbf{u}^*$ such that

$$\mathbf{u}^{*\top} \mathbf{A} = \mathbf{0}_R^\top, \quad (4)$$

$$\mathbf{u}^{*\top} \boldsymbol{\iota}_m = 0, \quad (5)$$

$$\mathbf{u}^{*\top} \mathbf{C} \mathbf{u}^* = \mathbf{v}^{*\top} \mathbf{C} \mathbf{v}^*, \quad (6)$$

- Under Assumptions AD and JK, there exists at least one vector $\mathbf{u}^*$ satisfying (4) - (6). Moreover, $\mathbf{u}^{*\top} \mathbf{C} \mathbf{v}^* = 0$. (Theorem 2.2)

Theorem 2.3: Assume $\mathcal{V} \cap \text{null}(C) = \emptyset$. Under Assumptions AD and JK,

$$\mathcal{J} := \frac{\tilde{\varphi} - \varphi}{\tilde{\sigma}} \xrightarrow{d} t_1,$$

as $N, T \rightarrow \infty$ and $N/T \rightarrow \gamma^2 \in (0, \infty)$, where

$$\tilde{\sigma}^2 := (\mathbf{u}^{*\top} \hat{\varphi})^2 \quad \text{and} \quad \tilde{\sigma} := \sqrt{\tilde{\sigma}^2}.$$

- Paves the way for hypothesis test, confidence intervals, and p -values...

Hypothesis tests. To test

$$H_0 : \varphi = \varphi_0 \quad \text{against} \quad H_A : \varphi \neq \varphi_0,$$

we form the jackknife t -statistic

$$\mathcal{J} := \frac{\tilde{\varphi} - \varphi_0}{\tilde{\sigma}}.$$

A level- α two-sided jackknife t -test rejects H_0 whenever $|\mathcal{J}| > t_{1,1-\alpha/2}$, where $t_{1,1-\alpha/2}$ is the $(1 - \alpha/2)$ quantile of a t -distribution with 1 degree of freedom. One-sided tests are obtained similarly.

p -values. For the two-sided test of $H_0 : \varphi = \varphi_0$ against $H_A : \varphi \neq \varphi_0$, the p -value associated with the test statistic on the previous slide is

$$p := 2(1 - F(|\mathcal{J}|)),$$

where $F(\cdot)$ denotes the CDF of the t -distribution with 1 degree of freedom. For one-sided alternatives $H_A : \varphi > \varphi_0$ and $H_A : \varphi < \varphi_0$, the corresponding p -values are

$$\begin{aligned} p^+ &:= 1 - F(\mathcal{J}) && \text{for } H_A : \varphi > \varphi_0, \\ p^- &:= F(\mathcal{J}) && \text{for } H_A : \varphi < \varphi_0. \end{aligned}$$

Confidence intervals. A $(1 - \alpha)$ confidence interval for φ is given by

$$\left[\tilde{\varphi} \pm t_{1,1-\alpha/2} \tilde{\sigma} \right],$$

where $t_{1,1-\alpha/2}$ is the $(1 - \alpha/2)$ quantile of the t -distribution with 1 degree of freedom. One-sided confidence intervals can be obtained similarly.

Example - continued

Recall

$$\hat{\varphi} = \begin{pmatrix} \hat{\varphi}^{(0)} \\ \hat{\varphi}^{(1)} \\ \hat{\varphi}^{(2)} \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \quad \text{and} \quad \mathbf{C} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix}.$$

The MVUJ weights vector is

$$\mathbf{v}^* = \left(2, -\frac{1}{2}, -\frac{1}{2} \right)^\top,$$

so that

$$\tilde{\varphi} = \mathbf{v}^{*\top} \hat{\varphi} = 2\hat{\varphi}^{(0)} - \frac{1}{2}\hat{\varphi}^{(1)} - \frac{1}{2}\hat{\varphi}^{(2)}.$$

As vector $\mathbf{u}^*$ satisfying (4) - (6) is

$$\mathbf{u}^* := \left(0, \frac{1}{2}, -\frac{1}{2} \right)^\top.$$

Hence the jackknife t -statistic is given by

$$\mathcal{J} = \frac{\tilde{\varphi} - \varphi}{\tilde{\sigma}} = \frac{4\hat{\varphi}^{(0)} - \hat{\varphi}^{(1)} - \hat{\varphi}^{(2)} - 2\varphi}{|\hat{\varphi}^{(1)} - \hat{\varphi}^{(2)}|} \xrightarrow{d} t_1.$$

Example - continued

- **Level- α two-sided jackknife t -test:** reject $H_0 : \varphi = \varphi_0$ whenever

$$\left| \frac{4\hat{\varphi}^{(0)} - \hat{\varphi}^{(1)} - \hat{\varphi}^{(2)} - 2\varphi_0}{|\hat{\varphi}^{(1)} - \hat{\varphi}^{(2)}|} \right| > t_{1,1-\alpha/2}.$$

- **two-sided jackknife p -value:**

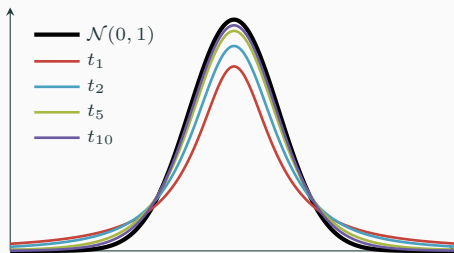
$$2 \left(1 - F \left(\left| \frac{4\hat{\varphi}^{(0)} - \hat{\varphi}^{(1)} - \hat{\varphi}^{(2)} - 2\varphi_0}{|\hat{\varphi}^{(1)} - \hat{\varphi}^{(2)}|} \right| \right) \right).$$

- **$(1 - \alpha)$ two-sided jackknife confidence interval:**

$$\left[2\hat{\varphi}^{(0)} - \frac{1}{2}\hat{\varphi}^{(1)} - \frac{1}{2}\hat{\varphi}^{(2)} \pm t_{1,1-\alpha/2} \cdot \frac{|\hat{\varphi}^{(1)} - \hat{\varphi}^{(2)}|}{2} \right].$$

Jackknife t_q -statistic

- Standard inference methods are based on a normal limiting distribution.
- Relative to these, our jackknife approach yields wider confidence intervals, and hypothesis tests with less power.
- We can, however, use additional subsamples to produce jackknife t_q -statistics.
- As $q \rightarrow \infty$, $t_q \rightarrow N(0, 1)$ very quickly:



Jackknife t_q -statistic

- Suppose there exist q vectors $\mathbf{u}_1^*, \dots, \mathbf{u}_q^*$ such that, writing

$$\mathbf{U}^* := (\mathbf{u}_1^*, \dots, \mathbf{u}_q^*),$$

the following conditions hold:

$$\mathbf{U}^{*\top} \mathbf{A} = \mathbf{0}_{q \times R}, \tag{7}$$

$$\mathbf{U}^{*\top} \boldsymbol{\nu}_m = \mathbf{0}_q, \tag{8}$$

$$\mathbf{U}^{*\top} \mathbf{C} \mathbf{U}^* = \mathbf{v}^{*\top} \mathbf{C} \mathbf{v}^* \mathbf{I}_q. \tag{9}$$

Theorem 2.4: Assume $\mathcal{V} \cap \text{null}(C) = \emptyset$. Let v^* solve (3), and let $u_1^*, \dots, u_q^*$ be such that with $U^* := (u_1^*, \dots, u_q^*)$ conditions (7) - (9) hold. Under Assumptions AD and JK,

$$\mathcal{J}_q := \frac{\tilde{\varphi} - \varphi}{\tilde{\sigma}_q} \xrightarrow{d} t_q, \quad (10)$$

as $N, T \rightarrow \infty$ and $N/T \rightarrow \gamma^2 \in (0, \infty)$, where

$$\tilde{\sigma}_q^2 := \frac{1}{q} \sum_{l=1}^q (u_l^{*\top} \hat{\varphi})^2, \quad \text{and} \quad \tilde{\sigma}_q := \sqrt{\tilde{\sigma}_q^2}.$$

Example - continued

Previously we considered $m = 3$ estimators. This does not give us enough degrees of freedom to construct anything beyond a t_1 :

- 1 df to remove bias
- 1 df for summing constraint
- 1 df for variance weights vector

We extend to further subsamples. Assume $T = 3T_3$ where T_3 is an integer. Consider subsamples

$$S_1 := \{1, \dots, N\} \times \{1, \dots, T_3\},$$

$$S_2 := \{1, \dots, N\} \times \{T_3 + 1, \dots, 2T_3\},$$

$$S_3 := \{1, \dots, N\} \times \{2T_3 + 1, \dots, T\}.$$

Let $\hat{\varphi}^{(j)}$ be the LS estimator using subsample S_j for $j = 1, 2, 3$. Collecting these with the full-sample estimator $\hat{\varphi}^{(0)}$ gives

$$\hat{\varphi} = (\hat{\varphi}^{(0)}, \hat{\varphi}^{(1)}, \hat{\varphi}^{(2)}, \hat{\varphi}^{(3)})^\top.$$

Example - continued

As previously we obtain,

$$\sqrt{NT}(\hat{\boldsymbol{\varphi}} - \boldsymbol{\varphi}_{\boldsymbol{\nu}_4}) = \mathbf{z}_{NT} + \mathbf{A}\boldsymbol{\mu}_{NT} + \mathcal{O}_p(1),$$

with $\mathbf{z}_{NT} \xrightarrow{d} \mathcal{N}(\mathbf{0}_4, \sigma^2 \mathbf{C})$,

$$\mathbf{A} = \begin{pmatrix} 1 \\ 3 \\ 3 \\ 3 \end{pmatrix}, \quad \text{and} \quad \mathbf{C} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & 0 & 0 \\ 1 & 0 & 3 & 0 \\ 1 & 0 & 0 & 3 \end{pmatrix}.$$

There is a unique MVUJ estimator associated with this design which is given by

$$\tilde{\varphi} = \frac{3}{2}\hat{\varphi}^{(0)} - \frac{1}{6}\hat{\varphi}^{(1)} - \frac{1}{6}\hat{\varphi}^{(2)} - \frac{1}{6}\hat{\varphi}^{(3)}.$$

Define

$$\mathbf{u}_1^* := \left(0, -\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, 0\right)^\top \quad \text{and} \quad \mathbf{u}_2^* := \left(0, -\sqrt{\frac{1}{18}}, -\sqrt{\frac{1}{18}}, \sqrt{\frac{2}{9}}\right)^\top.$$

Example - continued

Writing $U^* := (u_1^*, u_2^*)$,

$$U^{*\top} A = 0_2^\top, \quad U^{*\top} \iota_4 = 0_2, \quad \text{and} \quad U^{*\top} C U^* = v^{*\top} C v^* I_2.$$

The corresponding jackknife standard error is

$$\tilde{\sigma}_2^2 = \frac{1}{2} (u_1^{*\top} \hat{\varphi})^2 + \frac{1}{2} (u_2^{*\top} \hat{\varphi})^2,$$

and the jackknife t_2 -statistic is

$$\mathcal{J}_2 := \frac{\tilde{\varphi} - \varphi}{\tilde{\sigma}_2} \xrightarrow{d} t_2.$$

Jackknife t_q^2 -statistic

- So far we have only considered inference on a scalar parameter.
- It is possible to extend our method of inference to encompass vector parameters and to accommodate joint hypothesis tests.
- This is achieved by constructing a jackknife counterpart to Hotelling's t^2 -statistic (?):

$$\mathcal{J}_q^2 := \left((\mathbf{v}^{*\top} \otimes \mathbf{I}_k) \mathbf{h}(\hat{\boldsymbol{\varphi}}) \right)^\top \hat{\boldsymbol{\Sigma}}_q^{-1} \left((\mathbf{v}^{*\top} \otimes \mathbf{I}_k) \mathbf{h}(\hat{\boldsymbol{\varphi}}) \right). \quad (11)$$

Theorem 2.5: Assume $\mathcal{V} \cap \text{null}(\mathbf{C}) = \emptyset$. Let $\mathbf{v}^*$ solve (3), and let $\mathbf{u}_1^*, \dots, \mathbf{u}_q^*$ be such that with $\mathbf{U}^* := (\mathbf{u}_1^*, \dots, \mathbf{u}_q^*)$ conditions (7) - (9) hold. Further, assume that $q \geq k$, $\mathbf{G}(\varphi)\mathbf{\Sigma}\mathbf{G}(\varphi)^\top \succ \mathbf{0}_{k \times k}$, and $\|(\mathbf{A} \otimes \mathbf{I}_p)\boldsymbol{\mu}_{NT}\|_2 = \mathcal{O}(1)$. Under Assumptions AD and JK,

$$\mathcal{J}_q^2 \xrightarrow{d} t_{q,k}^2 \quad \text{if} \quad \mathbf{g}(\varphi) = \mathbf{0}_{k \times 1},$$

as $N, T \rightarrow \infty$ and $N/T \rightarrow \gamma^2 \in (0, \infty)$, where $t_{q,k}^2$ denotes Hotelling's t^2 -distribution with parameters q and k .

- $t_{q,k}^2 \rightarrow \chi_k^2$ as $q \rightarrow \infty$.
- $\frac{q-k+1}{kq} t_{q,k}^2 \sim F_{k, q-k+1}$.

Multi-dimensional Fixed Effects

- This talk has focused on one-way effects.
- The paper also discussed two-way effects in detail.
- The method can be extended to multi-dimensional models more generally.
- In order to do so, we need to know $\mathbf{A}$ and $\mathbf{C}$.

Multi-dimensional Fixed Effects

A can be reasoned heuristically:

- One-way models:

$$\frac{\text{dimension of fixed effects}}{\text{order of the standard error}} = \frac{N}{\sqrt{NT}} = \sqrt{\frac{N}{T}}.$$

- Two-way models:

$$\frac{\text{dimension of fixed effects}}{\text{order of the standard error}} = \frac{N + T}{\sqrt{NT}} = \sqrt{\frac{N}{T}} + \sqrt{\frac{T}{N}}.$$

- Three-way models:

$$\frac{\text{dimension of fixed effects}}{\text{order of the standard error}} = \frac{N + T + M}{\sqrt{NTM}} = \sqrt{\frac{N}{TM}} + \sqrt{\frac{T}{NM}} + \sqrt{\frac{M}{NT}}.$$

Multi-dimensional Fixed Effects

- Many standard estimators are asymptotically linear, meaning that the leading order term in their expansion has the form

$$z_{NT} = \frac{1}{a_{NT}} \sum_{(i,t) \in S} \nu_{it},$$

where a_{NT} denotes an appropriate normalising sequence and S denotes a set of indices.

- For example, when the object of interest is a common parameter one typically has

$$z_{NT} = \frac{1}{\sqrt{NT}} \sum_{(i,t) \in S_0} \nu_{it},$$

with $S_0 := \{1, \dots, N\} \times \{1, \dots, T\}$.

- If, for example, ν_{it} are iid, then C is determined by overlap between subsamples. This also holds more generally.

- Data are generated according to

$$y_{it} = \varphi x_{it} + \lambda_i + \varepsilon_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T,$$

$$x_{i1} = 0,$$

$$x_{it} = 1\{y_{i,t-1} > 0\}, \quad t \geq 2,$$

where $\varphi = 0.5$, $\lambda_i \sim \text{iid } \mathcal{N}(0, 1)$, and $\varepsilon_{it} \sim \text{iid } \mathcal{N}(0, 1)$.

- We study the LS estimator of φ , denoted by $\hat{\varphi}^{(0)}$.

- Assume $T = 2T_2$, $N = 2N_2$, and $N = 5N_5$ where T_2 , N_2 and N_5 are integers. Consider the following subsamples

$$S_1 := \{1, \dots, N\} \times \{1, \dots, T_2\},$$

$$S_2 := \{1, \dots, N\} \times \{T_2 + 1, \dots, T\},$$

$$S_3 := \{1, \dots, N_2\} \times \{1, \dots, T\},$$

$$S_4 := \{N_2 + 1, \dots, N\} \times \{1, \dots, T\},$$

$$S_{4+g} := \{(g-1)N_5 + 1, \dots, gN_5\} \times \{1, \dots, T\}, \quad g = 1, \dots, 5.$$

- Let $\hat{\varphi}^{(j)}$ denote the LS estimator computed on subsample S_j . We compare three different constructions.

- JK(a), utilises subsamples S_1 and S_2 , alongside S_0 . With $\hat{\varphi}_{(a)} := (\hat{\varphi}^{(0)}, \hat{\varphi}^{(1)}, \hat{\varphi}^{(2)})$.
- The MVUJ estimator is

$$\tilde{\varphi}_{(a)} := 2\hat{\varphi}^{(0)} - \frac{1}{2}\hat{\varphi}^{(1)} - \frac{1}{2}\hat{\varphi}^{(2)}.$$

- There is one admissible variance weights vector

$$\mathbf{u}^* = \left(0, \frac{1}{2}, -\frac{1}{2}\right)^\top.$$

- The jackknife t -statistic is

$$\mathcal{J}(a) := \frac{\tilde{\varphi}_{(a)} - \varphi}{\tilde{\sigma}_{(a)}} \xrightarrow{d} t_1 \quad \text{with} \quad \tilde{\sigma}_{(a)} = |\mathbf{u}^{*\top} \hat{\varphi}_{(a)}|.$$

Simulations I

- JK(b) uses additional cross-sectional partitions. In particular $\hat{\varphi}_{(b)} := (\hat{\varphi}^{(0)}, \hat{\varphi}^{(1)}, \hat{\varphi}^{(2)}, \hat{\varphi}^{(3)}, \hat{\varphi}^{(4)})^\top$.
- The MVUJ estimator is not unique, so we use a convenient construction

$$\tilde{\varphi}_{(b)} := \frac{2}{3}\hat{\varphi}^{(0)} - \frac{1}{2}\hat{\varphi}^{(1)} - \frac{1}{2}\hat{\varphi}^{(2)} + \frac{2}{3}\hat{\varphi}^{(3)} + \frac{2}{3}\hat{\varphi}^{(4)}.$$

- Two admissible variance weights vectors are

$$\mathbf{u}_1^* = \left(0, \frac{1}{2}, -\frac{1}{2}, 0, 0\right)^\top \quad \text{and} \quad \mathbf{u}_2^* = \left(0, 0, 0, \frac{1}{2}, -\frac{1}{2}\right)^\top.$$

- The jackknife t -statistic is

$$\mathcal{J}(b) := \frac{\tilde{\varphi}_{(b)} - \varphi}{\tilde{\sigma}_{(b)}} \xrightarrow{d} t_2 \quad \text{with} \quad \tilde{\sigma}_{(b)} := \sqrt{\frac{1}{2}(\mathbf{u}_1^{*\top} \tilde{\varphi}_{(b)})^2 + \frac{1}{2}(\mathbf{u}_2^{*\top} \tilde{\varphi}_{(b)})^2}$$

- JK(c) uses additional cross-sectional partitions. In particular,

$$\hat{\varphi}_{(c)} := (\hat{\varphi}^{(0)}, \hat{\varphi}^{(1)}, \hat{\varphi}^{(2)}, \hat{\varphi}^{(5)}, \hat{\varphi}^{(6)}, \hat{\varphi}^{(7)}, \hat{\varphi}^{(8)}, \hat{\varphi}^{(9)})^\top,$$

where $\hat{\varphi}^{(4+g)}$ is the LS estimator on the g -th cross-sectional fifth.

- The MVUJ estimator is not unique so we again use a convenient construction

$$\tilde{\varphi}_{(c)} := \hat{\varphi}^{(0)} - \frac{1}{2}\hat{\varphi}^{(1)} - \frac{1}{2}\hat{\varphi}^{(2)} + \frac{1}{5}\sum_{g=1}^5 \hat{\varphi}^{(4+g)}.$$

- Five admissible variance weights vectors are

$$\mathbf{u}_1^* = \left(0, \frac{1}{2}, -\frac{1}{2}, 0, 0, 0, 0, 0\right)^\top,$$

$$\mathbf{u}_2^* = \left(0, 0, 0, \frac{1}{\sqrt{10}}, -\frac{1}{\sqrt{10}}, 0, 0, 0\right)^\top,$$

$$\mathbf{u}_3^* = \left(0, 0, 0, \frac{1}{\sqrt{30}}, \frac{1}{\sqrt{30}}, -\frac{2}{\sqrt{30}}, 0, 0\right)^\top,$$

$$\mathbf{u}_4^* = \left(0, 0, 0, \frac{1}{\sqrt{60}}, \frac{1}{\sqrt{60}}, \frac{1}{\sqrt{60}}, -\frac{3}{\sqrt{60}}, 0\right)^\top,$$

$$\mathbf{u}_5^* = \left(0, 0, 0, \frac{1}{10}, \frac{1}{10}, \frac{1}{10}, \frac{1}{10}, -\frac{4}{10}\right)^\top.$$

- The jackknife t -statistic is

$$\mathcal{J}(c) := \frac{\tilde{\varphi}_{(c)} - \varphi}{\tilde{\sigma}_{(c)}} \xrightarrow{d} t_5 \quad \text{with} \quad \tilde{\sigma}_{(c)} := \sqrt{\frac{1}{5} \sum_{l=1}^5 \left(\mathbf{u}_l^{*\top} \hat{\varphi}_{(c)}\right)^2}.$$

Simulations I

N	T		LS	JK			HK			HJ		
				JK(a)	JK(b)	JK(c)	$h = 1$	$h = 2$	$h = 5$	$\ell = 1$	$\ell = 2$	$\ell = 5$
				t_1	t_2	t_5						
100	10	Bias	-0.1701	0.0150	0.0147	0.0150	-0.0445	-0.0955	-0.1737	-0.1696	-0.0496	-0.0170
		Std. Err.	0.0793	0.0956	0.0957	0.0958	0.0797	0.0842	0.0819	0.0794	0.0890	0.1016
		Coverage	0.4124	0.9538	0.9455	0.9286	0.9031	0.7469	0.3961	0.7094	0.9913	0.9103
		Length	0.3045	2.1164	0.7039	0.4162	0.3045	0.3045	0.3045	0.4075	0.5114	0.3443
250	20	Bias	-0.0910	0.0034	0.0033	0.0032	-0.0190	-0.0235	-0.0431	-0.0908	-0.0275	-0.0142
		Std. Err.	0.0365	0.0401	0.0401	0.0401	0.0364	0.0377	0.0377	0.0365	0.0391	0.0416
		Coverage	0.2862	0.9513	0.9442	0.9375	0.9158	0.8865	0.7590	0.4496	0.9902	0.9914
		Length	0.1403	0.8438	0.2962	0.1826	0.1403	0.1403	0.1403	0.1654	0.2110	0.2446
1000	80	Bias	-0.0245	0.0002	0.0002	0.0002	-0.0043	-0.0015	-0.0027	-0.0245	-0.0093	-0.0055
		Std. Err.	0.0091	0.0093	0.0093	0.0093	0.0091	0.0092	0.0092	0.0091	0.0092	0.0094
		Coverage	0.2322	0.9539	0.9512	0.9470	0.9205	0.9453	0.9371	0.2547	0.9029	0.9713
		Length	0.0355	0.1877	0.0696	0.0446	0.0355	0.0355	0.0355	0.0367	0.0398	0.0434

LS denotes the uncorrected LS estimator, JK(a)–JK(c) denote the jackknife estimators, HK denotes the analytical bias correction of ? with bandwidth h , and HJ denotes the moving block bootstrap of ? with block length ℓ .

Simulations II

- We consider a panel probit model with interactive fixed effects. Data are generated according to

$$y_{it} = 1 \{ \varphi x_{it} + \lambda_i \gamma_t + \varepsilon_{it} \geq 0 \}, \quad i = 1, \dots, N, \quad t = 1, \dots, T,$$

where $\varphi = 1$, and $\lambda_i \sim \text{iid } \mathcal{N}(0, 1)$, $\gamma_t \sim \text{iid } \mathcal{N}(0, 1)$, and $\varepsilon_{it} \sim \text{iid } \mathcal{N}(0, 1)$ are mutually independent.

- We consider two designs for the covariate x_{it} , exogenous and predetermined,

$$x_{it} = 1 \{ \eta_{it} > 0 \},$$

$$x_{it} = 1 \{ y_{i,t-1} > 0 \} + 1 \{ \eta_{it} > 0 \}, \quad x_{i1} = 1 \{ \eta_{i1} > 0 \},$$

where $\eta_{it} \sim \text{iid } \mathcal{N}(0, 1)$ and is independent of λ_i , γ_t , and ε_{it} .

- We study the ML estimator of φ , denoted by $\hat{\varphi}^{(0)}$.

- Assume $N = 2N_2$ and $T = 2T_2$, where N_2 and T_2 are integers. Our first jackknife estimator is based on the subsamples

$$S_1 := \{1, \dots, N\} \times \{1, \dots, T_2\},$$

$$S_2 := \{1, \dots, N\} \times \{T_2 + 1, \dots, T\},$$

$$S_3 := \{1, \dots, N_2\} \times \{1, \dots, T\},$$

$$S_4 := \{N_2 + 1, \dots, N\} \times \{1, \dots, T\}.$$

- Let $\hat{\varphi}^{(j)}$ denote the ML estimator computed on subsample S_j , and collect

$$\hat{\varphi}_{(a)} := (\hat{\varphi}^{(0)}, \hat{\varphi}^{(1)}, \hat{\varphi}^{(2)}, \hat{\varphi}^{(3)}, \hat{\varphi}^{(4)})^\top.$$

- JK(a) and JK(b) are both based on $\hat{\varphi}_{(a)}$. The MVUJ estimator is

$$\tilde{\varphi}_{(a)} := 3\hat{\varphi}^{(0)} - \frac{1}{2} \left(\hat{\varphi}^{(1)} + \hat{\varphi}^{(2)} + \hat{\varphi}^{(3)} + \hat{\varphi}^{(4)} \right).$$

- Two admissible variance weights vectors are

$$\mathbf{u}_1^* = \left(0, \frac{1}{2}, -\frac{1}{2}, 0, 0\right)^\top \quad \text{and} \quad \mathbf{u}_2^* = \left(0, 0, 0, \frac{1}{2}, -\frac{1}{2}\right)^\top.$$

- These yield two jackknife t -statistics

$$\mathcal{J}(a) := \frac{\tilde{\varphi}_{(a)} - \varphi}{\tilde{\sigma}_{(a)}} \xrightarrow{d} t_1 \quad \text{with} \quad \tilde{\sigma}_{(a)} = |\mathbf{u}_1^{*\top} \hat{\varphi}_{(a)}|,$$

$$\mathcal{J}(b) := \frac{\tilde{\varphi}_{(a)} - \varphi}{\tilde{\sigma}_{(b)}} \xrightarrow{d} t_2 \quad \text{with} \quad \tilde{\sigma}_{(b)} = \sqrt{\frac{1}{2}(\mathbf{u}_1^{*\top} \hat{\varphi}_{(a)})^2 + \frac{1}{2}(\mathbf{u}_2^{*\top} \hat{\varphi}_{(a)})^2}.$$

Simulations II

- JK(c) uses additional cross-sectional partitions. Assuming $N = 5N_5$ with N_5 an integer, it utilises S_1 , S_2 , and

$$S_{4+g} := \{(g-1)N_5 + 1, \dots, gN_5\} \times \{1, \dots, T\}, \quad g = 1, \dots, 5.$$

- Collecting the estimators,

$$\hat{\varphi}_{(c)} := (\hat{\varphi}^{(0)}, \hat{\varphi}^{(1)}, \hat{\varphi}^{(2)}, \hat{\varphi}^{(5)}, \hat{\varphi}^{(6)}, \hat{\varphi}^{(7)}, \hat{\varphi}^{(8)}, \hat{\varphi}^{(9)})^\top,$$

where $\hat{\varphi}^{(4+g)}$ is the ML estimator on the g -th cross-sectional fifth S_{4+g} .

- The MVUJ estimator is

$$\tilde{\varphi}_{(c)} := \frac{9}{4}\hat{\varphi}^{(0)} - \frac{1}{2}\hat{\varphi}^{(1)} - \frac{1}{2}\hat{\varphi}^{(2)} - \frac{1}{20} \sum_{g=1}^5 \hat{\varphi}^{(4+g)}.$$

- Five admissible variance weights vectors are

$$\mathbf{u}_1^* = \left(0, \frac{1}{2}, -\frac{1}{2}, 0, 0, 0, 0, 0\right)^\top,$$

$$\mathbf{u}_2^* = \left(0, 0, 0, \frac{1}{\sqrt{10}}, -\frac{1}{\sqrt{10}}, 0, 0, 0\right)^\top,$$

$$\mathbf{u}_3^* = \left(0, 0, 0, \frac{1}{\sqrt{30}}, \frac{1}{\sqrt{30}}, -\frac{2}{\sqrt{30}}, 0, 0\right)^\top,$$

$$\mathbf{u}_4^* = \left(0, 0, 0, \frac{1}{\sqrt{60}}, \frac{1}{\sqrt{60}}, \frac{1}{\sqrt{60}}, -\frac{3}{\sqrt{60}}, 0\right)^\top,$$

$$\mathbf{u}_5^* = \left(0, 0, 0, \frac{1}{10}, \frac{1}{10}, \frac{1}{10}, \frac{1}{10}, -\frac{4}{10}\right)^\top.$$

- The jackknife t -statistic is

$$\mathcal{J}(c) := \frac{\tilde{\varphi}_{(c)} - \varphi}{\tilde{\sigma}_{(c)}} \xrightarrow{d} t_5 \quad \text{with} \quad \tilde{\sigma}_{(c)} := \sqrt{\frac{1}{5} \sum_{l=1}^5 (\mathbf{u}_l^{*\top} \hat{\boldsymbol{\varphi}}_{(c)})^2}.$$

Simulations II

N	T		Exog.				Pred.			
			MLE	JK(a)	JK(b)	JK(c)	MLE	JK(a)	JK(b)	JK(c)
				t_1	t_2	t_5		t_1	t_2	t_5
100	10	Bias	0.1197	-0.0326	-0.0326	-0.0344	0.1002	-0.0556	-0.0556	-0.0573
		Std. Err.	0.0973	0.1012	0.1012	0.0992	0.1446	0.1884	0.1884	0.1680
		Coverage	—	0.9483	0.9585	0.9639	—	0.9192	0.9304	0.9173
		Length	—	1.9973	0.8602	0.5807	—	1.8011	0.8617	0.5922
250	20	Bias	0.0554	-0.0028	-0.0028	-0.0023	0.0612	-0.0023	-0.0023	-0.0026
		Std. Err.	0.0372	0.0370	0.0370	0.0369	0.0328	0.0435	0.0435	0.0386
		Coverage	—	0.9517	0.9568	0.9617	—	0.9470	0.9575	0.9573
		Length	—	0.7664	0.3055	0.1949	—	0.6068	0.2606	0.1619
1000	80	Bias	0.0139	-0.0004	-0.0004	-0.0003	0.0155	-0.0003	-0.0003	-0.0003
		Std. Err.	0.0085	0.0084	0.0084	0.0084	0.0063	0.0062	0.0062	0.0062
		Coverage	—	0.9465	0.9495	0.9535	—	0.9468	0.9517	0.9492
		Length	—	0.1717	0.0657	0.0421	—	0.1273	0.0490	0.0312

Exog. and Pred. denote the exogenous and predetermined regressor designs, MLE is the uncorrected ML estimator, and JK(a)–JK(c) are the jackknife estimators.

Conclusion

- Under large N , large T asymptotics fixed effects estimators are typically consistent but asymptotically biased which complicates inference.
- This paper develops a new method of inference which has several advantages compared to existing methods.
- This method is not specific to fixed effects models and has further applications.